

A temperature-free morphological observable that distinguishes non-local from down-gradient heat transport at a melting interface

Abdelrahman Saifelden¹

¹Alexandria Higher Institute of Engineering and Technology (AIET), Alexandria, Egypt

Abstract. Ice and rock scallops migrate downstream as they evolve, a motion classically tied to the phase lag between the wall heat flux and the interface topography: a melting surface migrates downstream when the flux maximum lies downstream of the trough, i.e. when the flux–topography phase lies in $(\pi/2, \pi)$ (Ashton & Kennedy 1972; Gilpin et al. 1980; Hanratty 1981). A local, memoryless, down-gradient (“K-theory”) closure carries no such lag. Freezing a sinusoidal interface $y(x) = \bar{y} + a \sin(Kx)$ and projecting the time-averaged wall flux onto in-phase (amplitude rate β) and quadrature (migration ω_{mig}) harmonics, every coefficient is solver-measured: the conduction amplitude rate is wavelength-independent ($\beta/a \sim K^{+0.13}$), falsifying the K^2 and Mullins–Sekerka $|k|$ ansätze; the in-phase excess is negative at every amplitude and drive (smoothing-limited); and the quadrature grows sub-kinematically ($E_{cos} \propto U^{\approx 0.5}$, friction-velocity controlled), giving a damped, downstream-migrating mode $s = -\beta + i\omega_{mig}$. Swapping only the heat-transport operator at fixed momentum drives the quadrature to machine zero (-1.2×10^{-18} for a uniform eddy diffusivity) against -1.1×10^{-4} for resolved advection. We then construct a temperature-free observable $I \equiv \tau \cdot c_{mig}/\lambda = |Im(s)|/(2\pi|Re(s)|)$ in which conductivity, latent heat, and basal temperature drop cancel, computable from interface morphology alone; the solver pins $I = O(0.3\text{--}0.9)$, and a regime-matched reading of Bushuk et al. (2019) gives $I_{obs} \approx 0.05\text{--}0.4$ with the predicted downstream sign. We give a field-test protocol and a self-validated reanalysis harness (recovers I to $< 3\%$); the field test is proposed, not performed here, as the raw $h(x, t)$ arrays are not openly available.

1 Introduction

Scalloped erosion/melt patterns are ubiquitous on soluble and meltable surfaces — cave walls, glacier soffits, ice-shelf bases, meteorites, and lab ice (Curl 1966; Blumberg & Curl 1974; Ashton & Kennedy 1972; Bushuk et al. 2019). Their wavelength is fluid-selected through a friction-Reynolds criterion $Re_* = u_* \lambda / \nu \approx 2200$. A robust but under-exploited observation is that the pattern **migrates downstream** as it evolves — maximum ablation sits downstream of the trough, at flow reattachment (Blumberg & Curl 1974; Gilpin, Hirata & Cheng 1980).

The migration direction and its phase-lag origin are established (Gilpin et al. 1980; Hanratty 1981): downstream migration corresponds to a flux–topography phase in $(\pi/2, \pi)$, and a local, memoryless, down-gradient closure carries no such phase. What is **new here** is not that fact but three things built on it: a phase-separated harmonic decomposition that returns the amplitude and migration rates as independent solver measurements; a direct operator-swap experiment that quantifies how completely a down-gradient closure discards the migration-carrying asymmetry; and a temperature-free morphological observable that makes the phase lag measurable from field geometry alone, without knowing the basal temperature drop or any material

constant. The non-local, direction-selecting structure the migration requires is exactly the structure (memory, backscatter) that a local, down-gradient closure cannot represent.

Contributions.

- 30 1. A **phase-separated** wall-flux harmonic decomposition that returns the corrected mode $s = -\beta + i\omega_{mig}$ as an *output*, not an ansatz, with the conduction amplitude rate measured wavelength-independent — falsifying the $K^2/|k|$ curvature-smoothing ansätze for the conduction baseline, generalised across amplitude and drive (§3–§4).
2. A **direct operator-swap measurement** quantifying how far down-gradient closures fall short of the known phase-lag requirement: swapping only the heat-transport operator collapses the migration signal to machine zero (§5).
- 35 3. A **temperature-free, constant-free morphological observable** $I = \tau \cdot c_{mig}/\lambda$, with an implemented, self-validated reanalysis harness and a regime-matched literature bound, that makes the phase lag measurable from field morphology alone (§6–§8).

2 Method — harmonic decomposition of the wall flux

Freeze $y(x) = \bar{y} + a \sin(Kx)$ ($K = 2\pi n_w/L_x$), time-average the per-column melt flux $m(x) = -\kappa \partial \theta / \partial y$, and project onto the

40 two corrugation harmonics:

$$E_{sin} = 2\langle e(x) \sin(Kx) \rangle \text{ (in-phase with the **shape** } \rightarrow \text{ amplitude change, } \beta) \quad E_{cos} = 2\langle e(x) \cos(Kx) \rangle \text{ (in **quadra-}** \\ \text{ture } \rightarrow \text{ pattern **migration**, } \omega_{mig})$$

E_{cos} is the Curl-1966 reattachment/lee signature: zero by symmetry unless the flow carries a direction. Two decompositions are run: the conduction baseline m_{cond} (flow off) for the $\beta \sim K^p$ scaling, and the flow-induced excess $e = m_{flow} - m_{cond}$

45 versus drive U . Freezing the interface removes boundary motion so the flux is uncontaminated; this is *why* the two coefficients separate cleanly, where a moving-boundary fit conflates them and returns unphysical (both-negative) coefficients.

3 Findings

quantity	measured	ansatz it kills
conduction $\beta/a \sim K^{+0.13}$	wavelength- independent	K^2 curvature and Mullins–Sekerka $ k $ falsified
in-phase flow excess E_{sin}	negative at every driven (a, U)	no autonomous $+\alpha a^{1/2}$ growth — smoothing-limited
quadrature E_{cos}	≈ 0 at $U = 0$, grows $\propto U^{0.5-0.8}$	migration is real and friction-velocity $(u_* \sim U^{1/2})$ controlled, not kinematic $(\propto U)$

Corrected mode: $s(K, U) = -\beta(K, U) + i \cdot \omega_{mig}(U)$, $Re(s) < 0$ always, $Im(s) \propto U^{1/2}$ — a damped, downstream-migrating pattern, not a growth–saturation balance.

Why this counts as a result, not a check. The corrected mode *fell out of* the decomposition; every coefficient is solver-measured; physical units are pinned independently by the Curl selection law without choosing λ by hand; and the final ratio is constant-free and falsifiable.

4 Generalisation across amplitude and drive

4.1 Amplitude ($a_0/\lambda \in [0.05, 0.40]$). The two structural verdicts are not amplitude artefacts: the driven in-phase excess is **negative at every amplitude** (smoothing throughout, magnitude growing monotonically), and the conduction exponent stays $p \in [-0.27, +0.66]$, far below the K^2 (+2) and $|k|$ (+1) ansätze at every amplitude. *Honest caveat:* strict K -independence ($|p| < 0.6$) and amplitude-flat β/a hold only in the signal-rich regime $a_0/\lambda \gtrsim 0.2$; at shallow amplitude the single-wavenumber conduction signal is at the noise floor (the $K^2/|k|$ falsification is robust regardless). (amplitude_generalization_scan.)

4.2 Drive (U to 6, beyond the swept $[1.5, 3.0]$). The genuine falsification risk is that strong-drive lee separation adds an *in-phase* component (an emergent growth term the weak-drive sweep would miss). It does not: E_{sin} is **negative at every** U out to $U = 6$, and the migration follows a **sub-kinematic** $E_{cos} \propto U^{+0.48}$ ($\approx 1/2$), far below the kinematic-advection alternative U^1 . *Honest caveat:* migration is non-monotone — it peaks near $U \approx 4.5$ and rolls over by $U = 6$ (the lee structure reaches a limiting form), which *reinforces* the sub-kinematic reading. (drive_window_scan.)

U	in-phase E_{sin}	quadrature E_{cos}
0.0	-5.17×10^{-6} (smoothing)	$+4.9 \times 10^{-8}$ (≈ 0 , parity control)
1.5	-6.23×10^{-5}	-6.59×10^{-5}
3.0	-9.19×10^{-5}	-1.14×10^{-4}
4.5	-8.86×10^{-5}	-1.38×10^{-4}
6.0	-4.81×10^{-5}	-1.19×10^{-4}

5 A direct operator-swap measurement of the closure shortfall

A local, memoryless, down-gradient eddy diffusivity acting on a sinusoid is symmetric under $x \rightarrow -x$: it carries no flow direction and can only return an in-phase (real, damping) response — $E_{cos} \equiv 0$. This is the closure-space statement of the classical phase-lag requirement (Gilpin et al. 1980; Hanratty 1981). The resolved advective flux measured $E_{cos} \neq 0$, so migration requires the flow-direction-selecting, non-local reattachment asymmetry that K-theory discards. The contribution here is to make the shortfall **quantitative**: advancing the **same** momentum field at the **same** drive ($U = 3$, same interface) but transporting heat with the K-theory flux instead of resolved advection measures exactly how far each closure falls below the resolved migration:

heat transport (same $U = 3$, same interface)	E_{cos}	reading
resolved advective $-u \cdot \nabla \theta$	-1.14×10^{-4}	migrates
uniform eddy diffusivity $\nabla \cdot (K \nabla \theta)$	-1.2×10^{-18}	machine zero ($\sim 10^{-14}$ of resolved, $std = 0$)
flow-aware Smagorinsky $K_{eddy}(S)$	$+2.0 \times 10^{-6}$	$\sim 57\times$ below resolved

75 Both closures still **smooth** ($E_{sin} \neq 0$): K-theory can damp but cannot migrate. GPU/CuPy controls (Tesla P100) confirm (a) parity — $E_{cos} \rightarrow +2.1 \times 10^{-6}$ at $U = 0$ vs -8.4×10^{-5} at $U = 3$ ($\sim 40\times$ smaller); and (b) the high- Re_* migration exponent tightens to 0.52 ($nx = 160$, four seeds, scatter $\sim 1\text{--}2\%$), on the friction-velocity $\frac{1}{2}$. (`scallop_ktheory_control.py`.)

This is the morphological analogue of the memory and backscatter signatures that a down-gradient closure omits in *flux* space: the closure’s blindness does not merely lose information in the model — it leaves a fingerprint carved into the ice.

80 6 A constant-free, ΔT -free field test

A Stefan-number-free dimensional bridge ($\rho L v = q$, $q = -k_{th}(\Delta T/L_0)\partial\theta/\partial y$, length anchored by the corrugation wavelength) gives $\tau \sim \lambda^2/\Delta T$ (amplitude e-fold) and $c_{mig} \sim \Delta T/\lambda$ (migration speed). In the product the constants k_{th} , $\rho_i L$ **and** ΔT **cancel exactly**:

85 $I \equiv \tau \cdot c_{mig}/\lambda = Im(s)/(2\pi|Re(s)|)$ — the dimensionless ratio of migration rate to amplitude rate, basal- ΔT -free.

The solver pins $I = O(0.3\text{--}0.9)$ (0.33, 0.68, 0.88 at $n_w = 8, 12, 16$ — wavelength-dependent by a factor ≈ 2.7 , an $O(1)$ morphological number rather than a universal constant).

Decoy warning (a methodological result). $\tau \propto \lambda^2/\Delta T$ *looks* like classic curvature/Mullins–Sekerka smoothing, but the λ^2 comes entirely from the conduction length $\leftrightarrow$ time anchoring L_0^2/κ , **not** from a curvature operator — β/a is K -independent
90 (§3). Fitting $\tau \propto \lambda^2$ in the field would *wrongly confirm* the very curvature ansatz the harmonic probe falsified; the only discriminator is the K -exponent of β/a (≈ 0 here vs $+2$ for curvature), which the dimensional scaling hides. Anyone analysing field scallops should measure the K -exponent, not fit $\tau \propto \lambda^2$.

Field-test protocol. On one scallop train measure λ (crest spacing), c_{mig} (downstream crest displacement between two visits), and τ (amplitude e-fold). Then:

outcome	reading
$I_{obs} \in [0.2, 1.0]$, c_{mig} downstream	consistent with $s = -\beta + i\omega_{mig}$
$c_{mig} \approx 0$ ($Im(s) = 0$)	parity-symmetric closure — K-theory not falsified on the ice (contradicts the solver)
c_{mig} upstream	new physics (e.g. depositional, not melt-limited)
$I_{obs} \gg 1$ or $\ll 0.1$	$\lambda \leftrightarrow u_*$ anchoring mismatch

Because I is ΔT -free, a failure cannot be excused by an unknown basal ΔT ; measuring $c_{mig} + \lambda$ alone pins ΔT , so τ becomes a prediction and the system is over-determined.

7 Literature grounding

- 100 – λ (**wavelength**) — **solid**. $Re_* = u_* \lambda / \nu$ selection is mainstream: Blumberg & Curl (1974) ≈ 2200 , Thomas (1979) ≈ 1000 , Hsu–Locher–Kennedy (1979) $\lambda = 3180\nu / u_*$, Thorsness–Hanratty (1979)/Hanratty (1981) band $3100\text{--}6300\nu / u_*$.
- c_{mig} **direction & mechanism** — **solid**. Downstream migration with maximum ablation downstream of the trough at reattachment: Curl (1966), Ashton & Kennedy (1972), Hsu et al. (1979), Gilpin et al. (1980). Gilpin et al.’s linear stability predicts downstream migration when the heat-flux/topography phase shift lies in $(\pi/2, \pi)$ — the same statement as $Im(s) \neq 0$.
- 105 – c_{mig} **magnitude and τ** — **sparse**. Cave-scallop surveys report λ only; migration speed and amplitude decay time are essentially never tabulated together for one train.

Two derived consistency relations with the turbulence closure (Paper 1). The migration quadrature $Im(s)$ that distinguishes a real closure is the imaginary part of a single complex transport admittance $Z = a_r + ia_i$, whose real part is the amplitude-rate / backscatter of the closure (Paper 1; Kraichnan 1976) and whose imaginary part is this scallop migration; a
110 down-gradient eddy diffusivity is the projection $Z \mapsto \max(Re Z, 0)$, which zeroes migration and backscatter *together*. Because the flux response is moreover causal, $Re Z(k)$ and $Im Z(k)$ are Kramers–Kronig (Hilbert) conjugates: any *scale-dependent* eddy viscosity — which every real closure (spectral eddy viscosity, Smagorinsky $\propto |S|$) is — **forces** a nonzero migration, so K-theory’s pairing of a scale-dependent $Re Z$ with $Im Z \equiv 0$ is acausal (an $O(1)$ Kramers–Kronig residual). The measured migration $I = |Im Z| / (2\pi |Re Z|)$ is therefore not an incidental fact but the causality-mandated shadow of the same operator
115 that carries backscatter in the turbulence closure — a derived link, verified on the admittance model (discrete Kramers–Kronig reconstruction of $Im Z$ from $Re Z$ to 5.3%).

8 The one open gate — and a regime-matched bound

Sign caveat. Real ice ripples *grow* ($Re(s) > 0$, the Ashton–Kennedy/Gilpin/Hanratty moving-boundary instability), whereas the frozen-interface excess here is decay-only ($Re(s) < 0$) by construction. The migration $Im(s)$ is the robustly shared, K-
120 theory-falsifying piece; matching the *growing* branch needs an interface-growth mechanism (a R  thlisberger opening) this smoothing-only solver does not contain.

The migration transfers to a moving boundary. We close the part of this gate the solver *can* reach. Seeding a single mode on the interface and advancing it under the Stefan melt feedback ($\rho Lv = q$), the co-evolving interface tracks a **coherent damped-migrating eigenmode**: $\log|Z|$ and $\arg(Z)$ of the fundamental shape mode are both linear in time ($R^2 \geq 0.91$ for

125 amplitude and phase, two seeds $\times$ two Stefan numbers). It **decays** ($Re(s) < 0$ at every St — the decay-only branch is *physical*, not an artefact of freezing) and **migrates downstream**, with $Im(s)$ tracking the frozen-probe quadrature E_{cos} (predicted $Im = E_{cos}/(aSt) \approx -0.43$ vs measured -0.40 at $St = 10^{-3}$, $\sim 5\text{--}10\%$). The observable $I = |Im|/(2\pi|Re|) \approx 0.15\text{--}0.17$ is **St -invariant** (the rate scales as $1/St$, so $Re \cdot St$ and $Im \cdot St$ are constant to $\sim 3\%$), confirming I is a well-defined, normalization-free property of the mode. The migration — the closure-distinguishing quadrature — therefore **survives interface motion**. The
 130 total moving-interface decay runs $\sim 1.75\times$ faster than the flow-off conduction estimate (turbulence adds effective smoothing), so this motion-consistent $I \approx 0.16$ sits *below* the frozen flow-off band and *closer to* the Bushuk field bound below. The *growing* branch ($Re(s) > 0$) is still out of reach here. (`scallop_moving_boundary_check.py`, `subglacial/moving_boundary_check.json`.)

A regime-matched real-data bound. Bushuk et al. (2019, *JFM* 873) track an evolving ice–water interface $h(x, t)$ at
 135 sub-mm/15 Hz through transition $\rightarrow$ equilibrium $\rightarrow$ **adjustment**. Their adjustment regime (after the drive was cut $U = 1.00 \rightarrow 0.16ms^{-1}$) **damps and migrates downstream simultaneously** — the exact analogue of the frozen probe ($s = -\beta + i\omega_{mig}$, $Re(s) < 0$, $Im(s) \neq 0$), so the sign caveat does not apply. Reading their measured kinematics ($c_{mig} = 0.11mmmin^{-1} = 1.83 \times 10^{-6}ms^{-1}$, observed $\lambda \approx 13cm$, amplitude e-fold $\tau \approx 1\text{--}3h$ by eye):

$I_{obs} \approx 0.05\text{--}0.15$ (point ≈ 0.10) at the observed λ ; $O(0.05\text{--}0.4)$ overall; **downstream sign**. Measured the *same*
 140 *way* as the field data — a moving boundary with total (turbulent) smoothing, via the identical $Re(s) = d/dt \ln a_k$, $Im(s) = d\phi_k/dt$ protocol — the solver gives the motion-consistent $I \approx 0.16$ (above), a factor ≈ 1.6 above the point estimate, versus the factor ≈ 3.3 against the frozen flow-off band $[0.33, 0.88]$. The motion-consistent comparison thus **reduces the tension to mild** — I_{obs} is $O(0.1\text{--}1)$, neither ≈ 0 nor $\gg 1$, with the predicted downstream sign — a consistency check, not a falsification.

145 **The pin.** The one missing input is Bushuk’s underlying $h(x, t)$ arrays, available from the authors on request (`mitchell.bushuk@noaa.gov`); the published JFM supplement is a parameter table only, not the interface time series, and no open data deposit exists. The §6 ratio is computed from morphology alone, and the reanalysis harness is **already implemented and self-validated**: `harmonic_mode_rate(x, t, H)` rFFTs each frame, tracks the dominant mode $a_k(t)e^{i\phi_k(t)}$, fits $Re(s) = d/dt \ln a_k$, $Im(s) = d\phi_k/dt$, and returns I plus the downstream sign; on a synthetic damped, downstream-migrating train with
 150 **known** ($Re(s), c_{mig}, \lambda$) it recovers all three to $< 2\%$ and I to $< 3\%$. It ingests Bushuk’s arrays directly — no further code, just the data. (`scallop_field_test.py`.)

9 Discussion and scope

This is a solver-verified, quantitative characterization of the closure: the corrected mode, the operator-swap shortfall, and the temperature-free ratio are all self-consistent and GPU-reproducible. It is **not yet validated against field scallops** — §8 is
 155 precisely why that test is cheap (no ΔT needed). The banked claims are the qualitative three — $Im(s) \rightarrow 0$ at $U = 0$, strictly sub-linear migration, exponent $\rightarrow \frac{1}{2}$ at high Re_* (central value 0.52) — robust to the $\sim \pm 0.1$ scatter from spin-up/seed/ U -grid.

The solver-side moving-boundary measurement (§8) is now done — the migration transfers to an actually-moving interface and I is motion-invariant — so what a field dataset would still pin is the migration exponent’s precise value and the field I_{obs} from real $h(x, t)$.

160 10 Reproduce

```
python glaciers/scallop_amplitude_harmonics.py # $3-$4; writes figures/56_*.json
python glaciers/scallop_ktheory_control.py # $5 keystone control (CPU; xp=cupy for GPU)
python glaciers/scallop_field_test.py # $6/$8; figures/58_*.json (harness self-check + Bushu
python glaciers/scallop_moving_boundary_check.py # $8 moving-boundary migration transfer + I
165 pytest glaciers/tests/test_scallop_amplitude_harmonics.py \
    glaciers/tests/test_scallop_ktheory_control.py \
    glaciers/tests/test_scallop_field_test.py
```

Code and data availability. All scallop-migration and harmonic-decomposition results regenerate from the scripts in the reproduce section; the field-test harness and its synthetic validation data are in the public repository at <https://github.com/abdh0saifelden2-debug/K>.

170 *Author contributions.* A. Saifelden conceived the study, performed all computations and analysis, and wrote the manuscript.

Competing interests. The author declares that there are no competing interests.

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